

Lecture Sheet: Probability Basics for Machine Learning

Target Audience: Beginners in Data Science and Machine Learning

Objective: Understand the mathematical framework of uncertainty to build better predictive models.

1 Why Probability in Machine Learning?

Machine Learning is essentially **making predictions under uncertainty**. Data is rarely perfect; it contains noise, missing values, and ambiguity. Probability provides the framework to quantify that uncertainty.

- **Classification:** "What is the probability this email is spam?" ($P(\text{Spam}|\text{Email})$)
- **Regression:** "What is the most likely house price given this square footage?"
- **Deep Learning:** Neural networks output probabilities (via Softmax) to make decisions.

2 Core Definitions

Before diving into formulas, we must speak the language.

- **Experiment:** An action with an uncertain outcome (e.g., flipping a coin).
- **Sample Space (S):** The set of *all* possible outcomes.
 - *Coin flip:* $S = \{Heads, Tails\}$
 - *Six-sided die:* $S = \{1, 2, 3, 4, 5, 6\}$
- **Event (E):** A subset of the sample space (a specific outcome or group of outcomes we care about).
 - *Rolling an even number:* $E = \{2, 4, 6\}$
- **Random Variable (X):** A variable that takes on numerical values determined by the outcome of a random phenomenon.

3 Fundamental Rules of Probability

3.1 Basic Probability Formula

For a finite sample space where all outcomes are equally likely:

$$P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

Constraint: Probabilities always range between 0 (impossible) and 1 (certain).

$$0 \leq P(E) \leq 1$$

3.2 The Sum Rule (Union)

Used when asking "What is the chance of A **OR** B happening?"

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- **Mutually Exclusive:** If A and B cannot happen at the same time (e.g., rolling a 2 and a 5 on a single die), then $P(A \cap B) = 0$, so $P(A \cup B) = P(A) + P(B)$.

3.3 The Product Rule (Intersection)

Used when asking "What is the chance of A **AND** B happening?"

$$P(A \cap B) = P(A) \cdot P(B|A)$$

- **Independence:** If outcome A does not affect outcome B (e.g., flipping a coin twice), then $P(B|A) = P(B)$.
- Formula becomes: $P(A \cap B) = P(A) \cdot P(B)$

4 Conditional Probability

This is arguably the most important concept for ML. It measures the probability of an event occurring given that another event has already occurred.

Notation: $P(A|B)$ means "Probability of A given B".

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Example: The Dataset Filter

Imagine a dataset of 100 emails.

- 40 are Spam (S).
- 10 contain the word "Free" (F).
- 8 are Spam AND contain the word "Free" ($S \cap F$).

If we see the word "Free" in an email, what is the probability it is Spam?

$$P(S|F) = \frac{P(S \cap F)}{P(F)} = \frac{8/100}{10/100} = \frac{8}{10} = 0.8$$

Interpretation: There is an 80% chance the email is spam if it contains the word "Free".

5 Bayes' Theorem

Bayes' Theorem allows us to update our beliefs when we get new evidence. It is the foundation of **Naive Bayes Classifiers**.

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

- $P(A|B)$: **Posterior** (What we want to calculate).
- $P(B|A)$: **Likelihood** (How likely is the evidence given the class?).
- $P(A)$: **Prior** (Our initial belief).
- $P(B)$: **Evidence** (Total probability of the evidence).

6 Distributions (The Shape of Data)

In ML, we fit our data to distributions to make predictions.

6.1 Discrete Distributions (Countable outcomes)

- **Bernoulli**: Single trial, two outcomes (0 or 1). Example: A user clicks an ad or doesn't.
- **Binomial**: Multiple Bernoulli trials. Example: How many heads in 10 coin flips?

6.2 Continuous Distributions (Infinite outcomes)

- **Normal (Gaussian) Distribution**: The famous "Bell Curve". Most natural phenomena (height, blood pressure, errors in regression) follow this.
 - Defined by Mean (μ) and Standard Deviation (σ).
 - **ML Relevance**: Many algorithms (Linear Regression, Gaussian Naive Bayes) assume data is normally distributed.

7 Expectation and Variance

These are summary statistics used to describe random variables.

7.1 Expectation ($E[X]$)

The "weighted average" or the long-run average value.

$$E[X] = \sum x \cdot P(x)$$

Example: A lottery ticket costs \$2. There is a 1% chance to win \$100, and 99% chance to win \$0.

$$E[X] = (100 \cdot 0.01) + (0 \cdot 0.99) = 1 + 0 = \$1$$

Interpretation: On average, the ticket is worth \$1, but you paid \$2. It is a bad deal.

7.2 Variance ($Var(X)$)

Measures how "spread out" the data is from the mean. High variance means high uncertainty.

$$Var(X) = E[(X - \mu)^2]$$

- **Standard Deviation (σ):** $\sqrt{Var(X)}$. This puts the spread back in the same units as the original data.

8 Practice Exercises

Exercise 1: Basic Dice

You roll a fair six-sided die.

1. What is the probability of rolling a 3?
2. What is the probability of rolling a number greater than 4?
3. What is the probability of rolling an even number OR a 5?

Exercise 2: Dependent Events

You have a bag with 3 Red marbles and 2 Blue marbles. You draw one marble, **do not put it back**, and then draw a second marble.

1. What is the probability that both marbles are Red?

Exercise 3: Bayes' Theorem (Medical Test)

- 1% of the population has a disease ($P(D) = 0.01$).
- The test is 90% accurate for sick people ($P(Pos|D) = 0.90$).
- The test produces false positives 5% of the time for healthy people ($P(Pos|Healthy) = 0.05$).

Question: If a patient tests positive, what is the actual probability they have the disease? ($P(D|Pos)$)

9 Answer Key

Exercise 1

1. $1/6$
2. Numbers > 4 are $\{5, 6\}$. Probability $= 2/6 = 1/3$.
3. Evens are $\{2, 4, 6\}$. The number 5 is odd. These are mutually exclusive.
 $P(\text{Even}) + P(5) = 3/6 + 1/6 = 4/6 = 2/3$.

Exercise 2

- Draw 1 (Red): $3/5$.
- Draw 2 (Red, given 1st was Red): Only 2 Red left, 4 total marbles left. Probability is $2/4$ or $1/2$.
- Total: $(3/5) \times (1/2) = 3/10 = 0.3$.

Exercise 3 Use Bayes' Theorem:

$$P(D|Pos) = \frac{P(Pos|D) \cdot P(D)}{P(Pos)}$$

1. **Numerator:** $0.90 \cdot 0.01 = 0.009$
2. **Denominator ($P(Pos)$):** We can test positive if we are Sick (0.009) OR if we are Healthy ($0.99 \cdot 0.05 = 0.0495$).
Total $P(Pos) = 0.009 + 0.0495 = 0.0585$.
3. **Result:** $0.009/0.0585 \approx \mathbf{0.153}$

Conclusion: Even with a positive test, there is only a $\sim 15.3\%$ chance the patient is actually sick because the disease is so rare.